

PRE-REGISTRATION — VERSION 2.0

Plant Architecture Inversion via Coherence Gradient Field Specification

Architectural Inversion of *Raphanus sativus* Produced by Pre-Specified Inversion
of the Hydrotropic and Phototropic Coherence Gradient Field Before Germination
Five-Scenario Two-Chamber Cone Membrane System — Operative Protocol v2.3

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Amendment note v2.0 (2026-03-18): This document supersedes Pre-Registration v1.0 (2026-03-13, DOI: 10.5281/zenodo.19040399). The primary prediction P-INVERSION-1 is unchanged. The experimental apparatus has been substantially upgraded from the agar-cup system to a two-chamber opaque PVC pipe system with a cone membrane signal separator, three substrate types, and one free UV condition. The number of experimental scenarios has been expanded from four to five. Scenario 3 in this document is the direct equivalent of the EXP chamber in v1.0. All predictions were updated and re-locked at the timestamp above before any experimental data exists. No experimental data has been collected under either the v1.0 or v2.0 protocol at the time of this upload.

THE PRE-REGISTERED CLAIM

The organism is not the output of a genome alone. It is the output of a genome operating within a coherence gradient field. Change the field before germination and the genome executes a different developmental program. A different architectural form emerges. Same genome. Different field. Different organism.

This pre-registration specifies: the coherence gradient field conditions that will produce architectural inversion in *Raphanus sativus* (radish), the precise architectural outcome predicted for each of five experimental scenarios, the scoring method, the control conditions required, the success criteria, and the reproducibility requirements — all locked before the experiment is run.

Primary prediction (P-INVERSION-1): A radish seed placed in Scenario 3 — hydroponic foam above, UV-A LED embedded in reticulated foam below, complete external light deprivation — will germinate with root growing upward and shoot growing downward. Architecture Inversion Score (AIS) = 2 confirmed at T = 72 h and maintained at T = 168 h. Scenario 1 must record AIS = 0 in the same run.

1. Background and Theoretical Basis

1.1. The Standard Model and Its Incompleteness

Contemporary biology operates on the model: **Genome** → **Organism**. The genome is treated as the blueprint. To change the organism, one changes the genome (CRISPR, directed evolution, synthetic biology). This model is incomplete.

The complete model is: **Genome** + **Coherence Gradient Field** → **Organism**.

The genome contains *build capacity* — the capacity to build every structure the organism has built across its evolutionary history, and potentially more. The coherence gradient field is the *instruction set* that specifies which programs to express, where, and toward which attractor the developmental program navigates.

The organism that emerges is the resolution of the genome's build capacity within the field's instruction set.

1.2. Plant Architecture as Field Navigation

Standard plant architecture — root growing downward, shoot growing upward — is not genetically fixed. It is the attractor the genome resolves to within the standard coherence gradient field: water and minerals below, light above, gravity present.

Each axis navigates toward its coherence gradient:

- **Root:** Navigates toward water and minerals (positive hydrotropism, MIZ1/MIZ2 pathway), darkness (negative phototropism via phot1/phot2), and gravity (positive gravitropism).
- **Shoot:** Navigates toward light (positive phototropism via phot1/phot2), CO₂, and warmth (negative gravitropism).

Standard architecture is the attractor produced when all gradients are coherent: root signals (water, minerals, darkness) point downward; shoot signals (light) point upward. Gravity confirms both directions.

1.3. The Signal Deprivation Principle

The central engineering condition of this experiment (Scenarios 2–5) is **complete external light deprivation**. The UV-A LED embedded within the foam substrate is the *only* photonic signal in the coherence field of each chamber.

When a germinating seedling develops in complete darkness it enters **skotomorphogenesis** — the developmental program for light-seeking under total light deprivation. In this state:

- Hypocotyl elongation is maximal.
- All photoreceptors are primed at maximum sensitivity.
- There is zero ambient photonic noise.

Under these conditions the absolute efficiency of UV-A as a phototropic driver is irrelevant. The embedded UV-A source is not competing against anything. The signal-to-noise ratio is effectively infinite. UV-A activates phototropins (phot1, phot2). In a completely dark chamber with a single directional UV-A source, a skotomorphogenic seedling responds to the only photonic input in its entire coherence field.

Scenario 1 is the explicit exception: UV-A is freely available in the S1 air space with zero penetration cost. This is the true normal control — normal conditions, normal architecture confirmed.

1.4. The Geometric Prediction

If plant architecture is a field-navigated attractor rather than a genetically fixed output, then:

Inverting the coherence gradient field before germination should produce an inverted architectural attractor.

Placing hydroponic substrate above the seed and UV-A embedded substrate below the seed — before germination begins — should cause the root to navigate upward (toward water, suppressing gravitropism via MIZ1-mediated amyloplast degradation) and the shoot to navigate downward (toward the only photonic signal in the field).

This prediction is geometrically derived from the attractor framework *before* the experiment is run. The conditions are specified from the geometry. The outcome is predicted from the conditions. The experiment confirms or falsifies the geometry.

1.5. Empirical Basis for Individual Mechanisms

All component mechanisms are independently established in the published literature. No mechanism claimed here is novel in isolation. The novel claim is their combination in a pre-specified five-scenario structured experiment producing simultaneous full architectural inversion from a locked prediction.

- **Hydrotropism overriding gravitropism, radish:** Takahashi et al. (2003), *Plant Physiology* 132: 805–810. Radish (*Raphanus sativus*) roots grow upward toward water. Mechanism: MIZ1-mediated amyloplast degradation suppresses gravity sensing. Confirmed in both radish and *Arabidopsis*.
- **MIZ1 suppresses gravitropism under drought:** Zhang et al. (2025), *PNAS* 122(20): e2427315122. Under drought, roots ignore gravity and grow toward water (including upward). MIZ1 suppresses PIN auxin transporter activity that drives gravitropism. Confirmed in laboratory, clinostat, and field

conditions. Direct relevance: in Scenario 3, the root faces a drought-like condition — all water is above, none below. MIZ1-mediated suppression of gravity sensing is the predicted operative mechanism.

- **Root negative phototropism to UV-A/blue:** Silva-Navas et al. (2016), *PNAS* 113(11): 3119–3124. Roots actively avoid blue and UV-A wavelengths. Mechanism: asymmetric auxin redistribution via phot1/phot2. In Scenario 3, this drives root away from UV-A below, adding a second mechanism pointing root upward.
- **Hydrotropism overrides root negative phototropism:** When water and light are in competition, the hydrotropic signal dominates when the water gradient is sufficient (Kaneyasu et al., 2007). In Scenario 3, root negative phototropism (away from UV-A below) and positive hydrotropism (toward water above) both point upward. No conflict. Both mechanisms are cooperative.
- **Shoot positive phototropism toward buried light:** Christie (2007), *Annual Review of Plant Biology* 58: 21–45. Shoot phototropism is always positive regardless of light direction. When light is below, the hypocotyl bends downward. Mediated by phot1/phot2. In Scenario 3, the only photonic signal is UV-A embedded in Type B foam below. The shoot bends toward it.
- **Root positive phototropism unmasked in microgravity:** Kiss et al. (2012), *American Journal of Botany* and NASA TROPI ISS experiments. In microgravity, Arabidopsis roots show positive phototropism toward blue light — a capacity completely masked by gravitropism on Earth. Direct framework relevance: the genome contains programs that are suppressed by the standard coherence gradient field and become expressed when a component of that field is removed or overridden.
- **Photosynthesis from geothermal light (buried source):** Beatty et al. (2005), *PNAS* 102(26): 9306–9310. Green sulfur bacteria at 2,500 m depth perform photosynthesis using geothermal light from below. Confirmed analogue: photonic navigation toward a buried non-solar source is a real biological phenomenon across kingdoms.
- **Radiotropism toward embedded radiation:** Dadachova et al. (2007), *PLoS ONE*. Melanized fungi at Chernobyl grow toward ionizing radiation and use it as an energy source. Confirmed analogue: organisms navigate toward and harvest embedded non-standard energy sources under field deprivation of standard sources.

2. Pre-Registered Predictions

All predictions below are locked at the timestamp of this document's upload to Zenodo. No post-hoc modifications will be made to predictions. Amendments, if any, will be registered as separate versioned documents with new DOIs.

2.1. Architecture Inversion Score Definition

The Architecture Inversion Score (AIS) is the primary quantitative outcome measure, defined as follows. Angles are measured from the vertical (straight down = 0°, straight up = 180°) using a protractor held against the transparent section of the chamber wall at each observation timepoint.

AIS	Morphological Criterion	Interpretation
2	Root angle > 135° AND shoot angle < 45° from vertical downward (both axes fully inverted).	Full architectural inversion. P-INVERSION-1 confirmed.
1	One axis inverted (> 90° from expected), other axis partially or not inverted.	Partial inversion. One gradient dominant, other insufficient.
0	Root angle < 45° (downward) AND shoot angle > 135° (upward).	Standard architecture. No inversion.
−1	Both axes in same unexpected direction (both up or both down).	Disorientation. Mechanism failure or contamination.

2.2. Primary Prediction: P-INVERSION-1

P-INVERSION-1 (Primary): In **Scenario 3** (S1: Type A hydroponic foam above; S2: Type B UV-only foam below; cone apex down; complete external light deprivation), *Raphanus sativus* will germinate and grow with:

- Root directed **upward** (angle $> 135^\circ$ from vertical downward) at $T = 72$ h.
- Measurable root penetration depth into S1 Type A hydroponic foam recorded at $T = 72$ h.
- Shoot directed **downward** (angle $< 45^\circ$ from vertical downward) at $T = 72$ h.
- Measurable shoot penetration depth into S2 Type B UV foam recorded at $T = 72$ h.
- Architecture Inversion Score **AIS = 2** at $T = 72$ h, maintained at $T = 120$ h and $T = 168$ h.

2.3. Scenario Predictions: P-INVERSION-2 through P-INVERSION-5

P-INVERSION-2 (Scenario 1 — True Normal Control): S1: UV-A LED free in air space (zero penetration cost). S2: Type A hydroponic foam. Cone apex up. *Expected:* AIS = 0 at all timepoints. Root grows downward into S2 Type A foam. Shoot grows upward into free UV-lit air without penetrating any foam barrier. Standard architecture confirmed. This scenario is the validity gate for every run — if AIS $\neq 0$ in Scenario 1, the run is invalid regardless of any other result.

P-INVERSION-3 (Scenario 2 — UV separated above, hydroponic below): S1: Type B UV-only foam (UV-A embedded, no hydroponic). S2: Type A hydroponic foam. Cone apex up. *Expected:* Root grows downward into S2 Type A. Shoot grows upward and penetrates S1 Type B foam to reach embedded UV-A. Normal architecture maintained but active shoot penetration upward through UV foam required. AIS = 0 at $T = 72$ h with measurable shoot penetration into S1 Type B. This scenario isolates the phototropic penetration component without hydrotropic competition. Provides the inversion pair baseline against Scenario 3.

P-INVERSION-4 (Scenario 4 — All signals above, void below): S1: Type C foam (hydroponic + UV-A co-located). Void in S2. Cone apex up. *Expected:* Shoot grows upward penetrating S1 Type C to reach all co-located signals simultaneously. Root grows downward into void or is indeterminate. AIS = 0 or indeterminate. Maximum coherent field above, no competing signal below.

P-INVERSION-5 (Scenario 5 — All signals below, void above): S2: Type C foam (hydroponic + UV-A co-located). Void in S1. Cone apex down. *Expected:* Full inversion. Shoot grows downward penetrating S2 Type C to reach all co-located signals. Root grows upward into void or is indeterminate. AIS = 2 at $T = 72$ h. Maximum coherent field below, no competing signal above. This is the strongest possible inversion signal (no competing attractor in opposite direction). Comparison of Scenario 5 (AIS = 2, no competition) vs Scenario 3 (AIS = 2, competing hydroponic above) tests the effect of competing signal on inversion stability.

3. Experimental Design — v2.3 Apparatus

3.1. Overview

Five scenarios are run simultaneously in the same environment from the same seed batch and same substrate preparation batch. The only variable between scenarios is the configuration of the coherence gradient field. All 5 scenarios use the same two-chamber cone membrane apparatus with the same UV-A LED product.

3.2. The Two-Chamber Cone Membrane System

Each experimental unit consists of two opaque PVC pipe sections (50 mm internal diameter, 150 mm length each) joined at a cone membrane signal separator.

The **cone membrane** is a plastic funnel wrapped in metallized Mylar film (aluminized emergency blanket), serving two barrier functions:

1. **Liquid barrier:** prevents hydroponic solution migrating between S1 (top) and S2 (bottom).
2. **UV and light barrier:** prevents UV-A from the substrate in one chamber crossing into the other, and prevents external light entering at the joint.

A hydroponic net cup seats at the cone apex. The germinated seed is placed in the net cup with radicle extending through the net cup mesh. The cone orientation (apex up or apex down) determines which chamber the radicle faces and which chamber the emerging hypocotyl faces.

Cone orientation rule: The apex always points toward the void or least active substrate. The wide opening always faces the most active substrate.

3.3. Substrate Types

Three substrate types and one free UV condition are used across the five scenarios:

- **Free UV condition (Scenario 1 S1 only):** UV-A LED strip mounted free on inner face of S1 end cap, facing downward into S1 air space. No foam substrate in S1. UV-A freely available throughout S1 air space. Zero penetration cost. No water or minerals.
- **Type A — Hydroponic foam, no UV:** Reticulated polyurethane foam 20 ppi, saturated with hydroponic mineral solution (1/4 to 1/2 strength). No UV-A LED present at any depth. Signals: water and minerals only.
- **Type B — UV-only foam, no hydroponic:** Reticulated polyurethane foam 20 ppi, dry or minimally moist (plain distilled water only, no nutrients), with UV-A LED strip embedded 10–20 mm depth from the seed-facing surface. Signals: UV-A radiation only. Seedling must physically penetrate foam to reach UV-A.
- **Type C — Combined foam, hydroponic + UV:** Reticulated polyurethane foam 20 ppi, saturated with hydroponic mineral solution, with UV-A LED strip embedded 10–20 mm depth from the seed-facing surface. Signals: water, minerals, and UV-A co-located. Seedling must penetrate foam to reach all signals.

3.4. The Five Scenarios

ID	S1 (top)	S2 (bottom)	Cone	Purpose and prediction
S1	Free UV in air	Type A hydroponic	Apex up	True normal control. AIS = 0 required. Validity gate.
S2	Type B (UV only)	Type A hydroponic	Apex up	UV separated above. Normal architecture + shoot penetrates Type B upward. AIS = 0. Inversion pair baseline.
S3	Type A hydroponic	Type B (UV only)	Apex dn	PRIMARY CLAIM. Full inverted field. AIS = 2 predicted. Root up into Type A. Shoot down penetrating Type B.
S4	Type C (all signals)	Empty void	Apex up	All signals above. Normal + maximum coherent field. AIS = 0.
S5	Empty void	Type C (all signals)	Apex dn	All signals below. Maximum inversion signal. AIS = 2 predicted. No competing attractor above.

3.5. Materials

Item	Cost (USD)	Source
<i>Raphanus sativus</i> seeds (Cherry Belle)	\$2–4	Garden store, Amazon
Reticulated polyurethane foam 20 ppi	\$5–10	Aquarium supply, Amazon
Hydroponic mineral solution (seedling dilution, 1/4–1/2 strength)	\$5–12	Garden centre, hydro supplier
Opaque PVC pipe 50 mm ID, 150 mm lengths ×10	\$10–20	Plumbing / hardware store
PVC end caps ×10	\$5–10	Plumbing / hardware store
UV-A LED strip ~365 nm	\$8–15	Amazon
Metallized Mylar film (emergency blanket)	\$1–3	Camping supply, dollar store
Plastic funnels (50 mm rim) ×5	\$3–6	Kitchen supply
Hydroponic net cups 16–25 mm ×5	\$1–3	Hydro supplier, Amazon
Aquarium silicone sealant	\$5–8	Aquarium store, hardware
Distilled water	\$1–2	Grocery store
Red-light torch (>650 nm)	\$5–15	Astronomy / photography
Total	\$51–108	

3.6. Hydroponic Mineral Solution Composition

Option A (from salts):

- 1 mM KNO₃
- 0.5 mM KH₂PO₄
- 0.2 mM MgSO₄
- pH 5.8–6.5, adjusted with dilute KOH if needed
- Prepared in distilled water

Option B (simplified): Commercially available balanced hydroponic nutrient solution at 1/4 to 1/2 manufacturer's recommended seedling dilution, prepared in distilled water. Adequate ionic content for hydrotropism signalling.

3.7. Seed Germination Protocol

Seeds are germinated on damp paper towel in a sealed ziplock bag in complete darkness at 20–24°C for 48 hours before placement. Target state at placement: radicle 3–8 mm, intact, no kinking. All germination and placement steps under red-light only (>650 nm) to preserve skotomorphogenic state. The seedling must not be exposed to white light, blue light, or UV at any point from germination through placement to T = 0.

3.8. Light Exclusion Protocol

Complete external light deprivation for Scenarios 2–5 is the load-bearing engineering constraint.

1. Use opaque black PVC pipe (preferred) OR wrap full exterior of each chamber unit in black opaque tape or black plastic wrap with no gaps.
2. All ventilation ports fitted with reticulated foam plug light traps (20–25 mm depth, 20 ppi). Test each port: shine torch at outer face of foam plug; confirm no light visible from interior side.
3. Cone membrane joint sealed with continuous aquarium silicone bead. Light-seal test after curing: place torch inside assembled chamber; check for light leaks at joint from outside.
4. At each observation timepoint: red-light only for all interior inspection. Maximum 2 minutes light exposure per chamber per timepoint. Re-wrap or re-enclose immediately after photography. Record duration of light exposure at each timepoint.

Scenario 1 S1 is exempt from external light exclusion — the free UV LED is the design condition of that chamber. Scenario 1 S2 should be opaque.

3.9. Observation Protocol

Timepoint	Observation
T = 24 h	Germination check. Record radicle length and direction per scenario. Photograph from side under red-light. Confirm UV-A LED active in all powered scenarios. Re-seal and re-wrap Scenarios 2–5 immediately.
T = 48 h	Directional growth assessment. Record root and shoot direction per scenario (up/down/horizontal/indeterminate). Measure penetration depth into substrate (mm). Preliminary AIS. Re-seal and re-wrap.
T = 72 h	Primary outcome point. Measure root and shoot angles with protractor. Measure substrate penetration depth (mm). Record whether root or shoot tip has reached LED embedding depth in Type B or Type C foam. Calculate AIS for all scenarios. Photograph from side and above with scale ruler. Re-seal and re-wrap.
T = 120 h	Secondary confirmation. AIS re-measured. Stability of inversion assessed. Note any gravitropic correction (inversion at T = 72 but recovered at T = 120).
T = 168 h	Final outcome. Photograph all scenarios together on white background. Side-by-side comparison of Scenario 3 vs Scenario 1. Record final AIS for all scenarios.

4. Success Criteria and Falsification

4.1. Confirmation of P-INVERSION-1

P-INVERSION-1 is confirmed if and only if all of the following are met in the same experimental run:

1. Scenario 3: AIS = 2 at T = 72 h **and** maintained at T = 168 h.
2. Scenario 3: measurable shoot penetration into S2 Type B UV foam recorded at T = 72 h. Directional growth alone without foam penetration does not confirm active navigation toward the UV-A signal.
3. Scenario 1: AIS = 0 at all timepoints. If Scenario 1 records AIS ≠ 0, the run is invalid regardless of Scenario 3 result.
4. Complete light exclusion confirmed and logged for Scenarios 2–5 at every timepoint. Runs with unlogged or compromised light exclusion must be reported as flagged and analysed separately from primary results.

Reproducibility threshold: P-INVERSION-1 requires confirmation across **minimum 3 independent seeds** in **minimum 2 independent experimental runs** from separate seed batches before the result is considered reportable as confirmed.

4.2. Qualified Results

Gravitropic correction (AIS = 2 at T = 72, AIS = 0 at T = 120): This is a qualified result, not a failure. It confirms the coherence field inversion succeeded in redirecting architecture during the developmental window and that gravitropic correction occurs after window closure. Report full angle data at all timepoints. This is data about the competition between the coherence gradient field and gravitropism over developmental time.

AIS = 1 in Scenario 3: Partial inversion. One axis inverted, one insufficient. Not a confirmation of P-INVERSION-1. Report as qualified result. Investigate whether signal strength (LED embedding depth, intensity) or foam moisture level requires adjustment.

4.3. Falsification Criteria

The geometric prediction is falsified if, across minimum 3 seeds in 2 independent runs with confirmed light exclusion and confirmed Scenario 1 AIS = 0:

- Scenario 3 AIS = 0 at T = 72 h in all runs (gravity dominates over both combined coherence gradients at this field strength).
- Scenario 3 AIS = -1 in all runs (developmental confusion rather than inversion).

A single run failure does not falsify the prediction. Falsification requires consistent failure across the minimum reproducibility threshold with all controls confirmed valid.

5. Reproducibility Requirements

For any replication of this experiment to be considered valid and comparable to this pre-registration, the following minimum documentation must be provided:

1. Species, cultivar, and seed source (supplier, batch number, viability test result).
2. Substrate specification: foam ppi rating, supplier; nutrient solution product/formulation and dilution ratio; LED product name, stated wavelength (nm), embedding depth (mm from seed-facing surface), moisture condition of Type B foam.
3. Chamber specification: pipe material, inner diameter, section length; cone membrane material and construction; net cup dimensions.
4. Light exclusion method and test result for each chamber at T = 0 (confirmed light-tight Y/N per chamber).
5. Red-light torch wavelength and confirmation that white/blue/UV light was not used during any observation.
6. Temperature throughout experiment.
7. All 5 scenarios run simultaneously from T = 0 in the same environment.
8. AIS scores and substrate penetration depths (mm) at T = 72 h, T = 120 h, T = 168 h for all 5 scenarios.
9. UV-A LED active status confirmed at each timepoint for all powered scenarios.
10. Duration of light exposure per chamber per timepoint recorded.
11. Minimum one photograph per scenario per timepoint, labelled with scenario ID, timepoint, and date/time.
12. Any interventions or failure events recorded with timestamps and data-compromise flags.

6. The Comparison Structure

The five-scenario design is not five independent experiments. It is a structured reasoning space over two variables — UV direction and signal separation — that makes the primary result (Scenario 3) unambiguous by eliminating alternative explanations.

- **S1 vs S2:** UV free in air (zero penetration cost) vs UV in Type B foam above (high penetration cost). Same hydroponic below. Tests penetration cost alone.
- **S2 vs S3:** Primary inversion pair. Structurally identical — same substrate types, same signal strengths, same penetration requirements. Only direction flips. Cone flips with it. If architecture flips, the field is the author.
- **S3 vs S5:** Inversion with competing hydroponic above vs inversion with nothing above. Tests signal competition in the inversion result.
- **S4 vs S5:** All signals above vs all signals below. Maximum coherent field. Tests whether full inversion occurs when the signal is unambiguous and uncontested.

7. Relationship to the Broader Framework

This experiment is one component of a larger pre-registered research program (OrganismCore, DOI: [10.5281/zenodo.1898679](https://doi.org/10.5281/zenodo.1898679)) testing the geometric claim that organisms are the output of genomes operating within coherence gradient fields, and that changing the field changes the organism.

The plant inversion experiment is the first *prospective* test of this claim — a pre-specified prediction confirmed by experiment rather than a retrospective explanation of already-observed phenomena.

The retrospective confirmation record (confirmed cases across multiple kingdoms including hydrotropism overriding gravitropism in radish, root positive phototropism in microgravity, photosynthesis from geothermal buried light, and radiotropism toward embedded radiation sources, documented in the OrganismCore repository) is available at: github.com/Eric-Robert-Lawson/attractor-oncology

The oncological application of the coherence gradient field framework — that cancer represents a breakdown in coherence field maintenance within a tissue, with cells navigating to the wrong attractor because the local field has been disrupted — depends on establishing the principle that the field authors the form at the organismal level. The plant inversion experiment is the foundational proof of that principle.

8. Declaration

This document is submitted as a pre-registration. The experiment described has **not** been run at the time of this document's creation and upload. All predictions are made before any experimental data exists. Results, when obtained, will be reported in a separate document linked to this pre-registration by DOI. No modifications will be made to this document's predictions after upload. Amendments will be versioned and separately registered with new DOIs.

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The field came first. The genome was always listening. The architecture was never fixed. The inversion was always possible.
github.com/Eric-Robert-Lawson/attractor-oncology · doi.org/10.5281/zenodo.18986790 · ORCID: [0009-0002-0414-6544](https://orcid.org/0009-0002-0414-6544)